DevOps Overview

History, principles, CI/CD and common tools

DevOps Foundations

1. What is DevOps?

DevOps is a way of working that brings development and operations closer together. It combines collaboration, automation, continuous integration, delivery, monitoring and feedback.

The goal is to make software delivery faster and more reliable while reducing manual work and improving communication between teams.

Key ideas:

- Shared responsibility across development and operations
- Automation of repetitive tasks
- Fast feedback and continuous improvement
- Reliable and repeatable delivery processes

2. From Waterfall to DevOps

Waterfall used a mostly sequential process where requirements, development, testing and deployment happened in separate phases. Changes later in the process could be expensive or slow.

Agile introduced shorter development cycles, frequent releases and regular customer feedback. However, development and operations could still remain separated.

DevOps extended the focus on collaboration and fast feedback across the complete software delivery lifecycle.

Approach	Main focus
Waterfall	Sequential phases and planned releases
Agile	Iterative development and customer feedback
DevOps	Collaboration, automation and continuous delivery

3. The Three Ways of DevOps

Flow: Improve the movement of work from development through deployment to the customer. Reduce bottlenecks and unnecessary waiting.

Feedback: Create fast feedback loops using testing, monitoring, alerts and reviews so problems can be found early.

Continual Learning: Learn from failures and successes, experiment with improvements and keep improving the process.

4. CI/CD Pipeline

CI/CD automates important parts of the software delivery process. Continuous Integration validates changes frequently, while Continuous Delivery keeps software ready for release. Continuous Deployment can automatically deploy approved changes.

Typical lifecycle:

$$\text{Plan} \rightarrow \text{Code} \rightarrow \text{Build} \rightarrow \text{Test} \rightarrow \text{Release} \rightarrow \text{Deploy} \rightarrow \text{Operate} \rightarrow \text{Monitor} \rightarrow \text{Feedback}$$

5. Tools Used in DevOps

Area	Examples	Use
Source control	Git, GitHub	Manage and review code changes
CI/CD	Jenkins, GitHub Actions	Automate builds, tests and delivery
Containers	Docker, Kubernetes	Package and manage applications
IaC / Automation	Terraform, Ansible	Provision and configure infrastructure
Monitoring	Prometheus, Grafana	Collect metrics and visualize system health

6. Benefits of DevOps

- Faster and more frequent software releases
- Earlier detection of bugs and deployment issues
- Less repetitive manual work
- More consistent development and deployment processes
- Better communication between teams
- Faster response and recovery when problems occur
- Continuous improvement based on real feedback

7. Conclusion

DevOps is more than a set of tools. It combines people, processes and technology to improve the complete software delivery lifecycle.

The main ideas are improving flow, creating fast feedback loops and continuously learning. CI/CD, containers, infrastructure automation and monitoring provide technical support for these principles.

Thank you